

Attention-Based Graph Neural Network for Survival Prediction from Single-Cell Spatial Proteomics

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Abstract

Despite advances in cancer treatment, including surgery, radiotherapy, chemotherapy, and immunotherapy patient survival, cancer progression, therapeutic response remain suboptimal due to the limitations of traditional clinical and histopathological data that often overlook the spatial complexity of the tumor microenvironment (TME). Recent progress in spatial proteomics such as imaging mass cytometry (IMC), multiplex immunofluorescence (mIF), and other high-plex platforms enables high-resolution mapping of protein expression within intact tissues, but most studies still focus on individual biomarkers or predefined cell types, leaving gap in translating this rich spatial data into patient-level insights. To address this gap, we propose a data-driven computational pipeline that integrates a graph attention network (GAT) to model cell-type specific interactions and aggregates learning outcomes into a patient-level representation for survival analysis. This methodology is motivated by recent evidence that spatially analyzed proteins and transcriptomic signatures such as immune cells localization and compartment specific protein expression are strongly associated with survival and therapy response, yet remain underutilized in clinical decision-making. Our approach aims to deliver validated spatial biomarkers that improve risk stratification, guide personalized therapy, and address the unmet need for clinically actionable prognostic tools in cancer.

1. Introduction

Cancer prognosis has traditionally relied on histopathological grading and molecular subtyping, yet these approaches incompletely capture the spatial context of the tumor microenvironment (TME), which is now recognized as a key determinant of disease progression and response to therapy.[1, 2] Spatial proteomics has revolutionized our ability to detect and quantify proteins within the native architecture of tissues at single-cell resolution, providing unprecedented insight into the cellular interactions. Rather than viewing cancer as a homogeneous mass, spatial proteomics reveals it as a complex ecosystem where the organization and interplay of diverse cell types: cancer, immune, and stromal that critically influence disease progression, immune evasion, and therapeutic response.[3, 4] Recent advances in single-cell spatial proteomics, including IMC, multiplex immunohistochemistry/mIF, and ultra-high-plex digital spatial profiling,[1, 2, 5] now enable simultaneous measurement of tens to hundreds of proteins within intact tissue sections at near single-cell resolution. Despite these advances, analytical methods often summarize spatial proteomics into bulk-like or low-dimensional features[6], such as cell densities, pairwise neighbor counts, or simple spatial statistics. While these strategies can detect some prognostic patterns, they may miss complex, higher-order, and context-dependent cellular interactions.

The spatial arrangement of cells within the TME is increasingly recognized as a key determinant of patient prognosis and response to therapy.[2] Quantitative single-cell spatial analyses have shown that specific tumor-immune architectures, such as cell-type mixing, can predict survival outcomes.[4] Despite the introduction of novel therapies, many cancers continue to have poor outcomes, but spatial profiling has identified immune proteins whose localization within tumors correlates with better or worse survival.[7] Predictive models that explicitly encode these spatial relationships can enhance risk stratification, guide decisions about adjuvant therapy or immunotherapy, and highlight microenvironmental niches that represent potential therapeutic targets.

While spatial proteomics generates highly detailed data, most computational models fail to exploit the spatial dimension fully, often treating measurements as non-spatial features and as a result limiting both predictive performance and biological insight.[6, 8, 9] Studies have shown that explicitly modeling spatial features that takes up the concepts from ecology and epidemiology improves survival prediction[3] and reveals novel cell-cell interactions relevant to outcome. Recent studies showed that using hand-crafted spatial statistics like Ripley's K function, neighbor counts,

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and simple neighborhood matrices[3, 4] improve prediction but may not capture complex, multi-cellular patterns. Also these methods still rely on predefined feature sets and static aggregation schemes. In contrast attention-based graph networks can operate directly on spatial cell graphs, propagating information along edges that represent cell-cell proximity or interaction, but most existing applications use simple graph convolutional networks (GCNs) without attention mechanisms or direct coupling to survival models.[1, 10, 11]

This study bridges spatial proteomics and computational modeling to uncover how diseases progress by analyzing the spatial organization of cells in tissue. High-resolution mapping of proteins within the TME[12] reveals cell interactions,[3, 4] functional states, and region-specific markers linked to survival[8] and drug response,[13] highlighting the potential for developing new prognostic and biomarker tools. By integrating graph-based representation learning with attention mechanisms[1, 10, 9] our study aims to advance computational pathology toward precision oncology.

2. Related works

This methodology is motivated by recent evidence that spatially analyzed proteins and transcriptomic signatures, such as immune cell localization and compartment-specific protein expression, are strongly associated with survival and therapy response, yet remain underutilized in clinical decision-making.

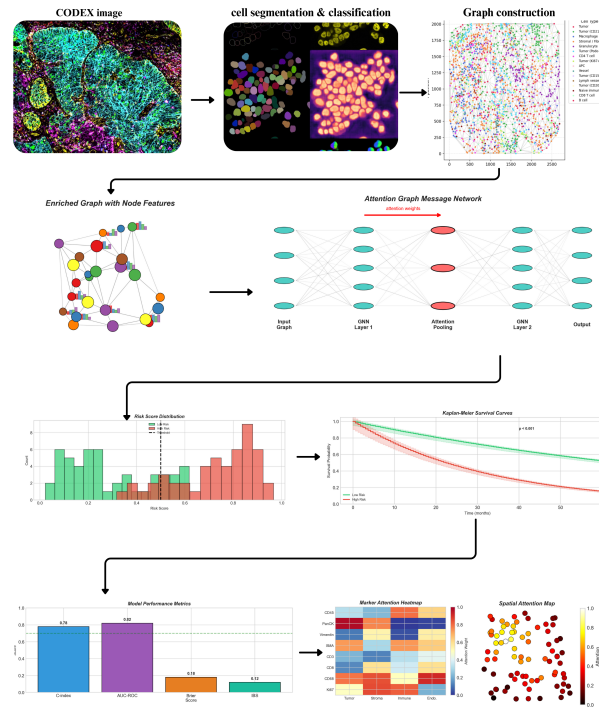
2.1. Spatial Proteomics Features

Spatial proteomics is an approach that integrates mass spectrometry along with advanced imaging techniques to locate protein information within cells in a high-resolution manner, while multiplex imaging measures the expression of multiple proteins within a single tissue without affecting its spatial position. [14, 15, 16] Though spatial proteomics can detect a limited number of proteins, it gives extensive spatial information with single-cell resolution.[17, 18] Moreover, the functional status of cells in the TME depends heavily on localization and spatial relationships. Multiplex imaging is crucial when it comes to spatial insights and interactions with TME. Currently, various proteomic imaging techniques such as IMC, MIBI-TOF, CyCIF, CODEX, and MELC[19] have been developed, and among them, MIBI and IMC can detect more than 40 targets via one-shot staining.[14, 17] In cancer research, multiplex imaging has been rapidly developed in recent years, and many more multiplex imaging technologies, along with spatial analysis methods, are emerging. [14] Multiplex immunostaining techniques, which can detect several dozens of different antigen molecules on a single formalin-fixed paraffin-embedded(FFPE) tissue has been developed and is being used in studies of clinical cohorts due to their easier application compared to immunohistochemistry (IHC). [14] One of the other clinical applications of this technique can be biomarker development, along with detecting combinations of multiple molecules at once, which would increase the prediction accuracy of therapy or prognosis.[14] Spatial analysis of the positional relationship of cells and cell types near a particular cell can provide better analysis of TME, and thus spatial analysis can be used in biomarker discovery. [14]

2.2. Spatial Graphs and Microenvironment Modeling

Spatial graph and microenvironment modeling are used in making raw multiplex imaging data into a structured representation of TME. Spatial graph represents the complex architecture of tissue by transforming multiplexed imaging into a geometrical structure[1].[20] Even though multiplex imaging allows molecular profiling of cellular environments, identifying disease from these rich datasets is difficult[1]. So, researchers use a graph neural network that leverages spatial protein profiles in tissue specimens and models tumor microenvironments as local subgraphs[1]. This spatial cellular graph has been applied to identify the spatial motifs of cancer recurrence, and this graph was more accurate in predicting outcome than the used deep learning models[20]. This approach uses cells as nodes and spatial interactions as edges.

TMEs are complex niches that are characterized by cellular and molecular heterogeneity. When it comes to tumor growth, metastasis, or even therapeutic resistance, TME components play a crucial role, and understanding the molecular mechanism can help in making better therapeutic strategies. Because of that, tumor pathology largely depends on the spatial organization of TME. When TME is modeled as a local subgraph, researchers discovered spatial mixing of specific tumor subtypes. In a clinical context, using spatial graphs with TME knowledge gives more accuracy in cancer recurrence, survival, and prediction[1].



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Figure 1: a. Multiplexed image pre-processing and spatial graph construction. b. Trajectory inference and pseudospace. c. survival prediction models

2.3. Attention-Based GNN Models

In single-cell spatial proteomics, attention-based GNN Models are unexplored. The integration of GNN with attention mechanisms creates a lot of broad scopes in terms of spatial variations, due to its inductive nature, which helps survival prediction. Recent frameworks like PathoGraph [9] and SurvGraph [10] leverage multi-head attention to stabilize the learning process and capture diverse biological signals. In Xiong et al. 2019, they leveraged the graph attention mechanism to integrate latent features from different drug representations[21]. Graph attention-based deep learning frameworks are also incorporated with multi-omics work[22] and spatial transcriptomics works to get spatial information and gene expression information. So working with spatial proteomics can provide new insights with an attention-based graph model

3. Proposed Method

The pipeline begins with collecting high-plex spatial proteomics data from tumor tissue images (cite7) then performing single-cell segmentation to extract cell protein profiles and spatial coordinates (cite 17,18) to construct spatial graphs where single cells are nodes annotated with cell-types and edges encode spatial proximity. A graph attention network (GAT) is then applied to learn the cell-cell interactions that represent the progression states, which are later aggregated into patient-level representation (cite 2,5) to put into the survival model (cite 3) as input for end-to-end risk stratification, validating the model's performance against clinical outcomes.

3.1. Dataset

For this research, we will utilize a publicly available HNSCC(Head and Neck Squamous Cell Carcinoma) dataset[3]. This dataset is a 39-plex spatial proteomics CODEX UPMC_HNC dataset from 81 HNSCC patients. This dataset included 308 image samples collected from 7 batches/coverslips, with between 1 and 8 samples belonging to each patient[3][1]. For the initial pipeline, we will be using the computed data from where we get samples annotated

with clinical data, including patient status, patient survival length, and HPV(Human papillomavirus) status. The computed data also include cell x-y locations, protein biomarkers(40 biomarkers), and spatial features(neighborhood matrix, computation result of Ripley-K functions). The protein markers include immune cell types (CD4 T cell , CD8 T cell , B cell), Tumor cell types(Ki67+, cd21+.CD20+, eg.) and stromal cells

3.2. Preprocessing

Raw protein intensity values have extreme outliers and a non-normal distribution. Several types of image processing can be initiated by cell segmentation using DeepCell, a neural network-based cell segmentation tool to segment cell boundaries and extract mean protein intensity per channel per cell to transform the raw image into actionable data. Normalize the protein expression value using quantile normalization and inverse hyperbolic sine transformation. Apply clustering (K- nearest neighbor) to annotate with cell types from the normalized expression vector[3][1].

3.3. Spatial Graph Construction & feature engineering

Graph construction transforms the preprocessed single-cell data into graph representation suitable for graph representation learning. Here each cell becomes a node and spatial relationships define edges. Formally, we construct a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ where:

- $\mathcal{V} = \{v_1, v_2, \dots, v_N\}$ is the set of nodes (cells),
- $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ is the set of edges (cell-cell connections).

Each node v_i is associated with a feature vector $\mathbf{h}_i \in \mathbb{R}^F$ representing the cell's protein expression profile and engineered features. Each cell is connected to its K spatially closed neighbors; this approach ensures uniform node degree regardless of local cell density. Feature engineering enriches the graph representation by computing additional nodes with the initial node feature vector, which consisted of normalized protein expression and edge features that captured biological and topological information. We augment node features with features like cell type assignment, Local cell density, neighborhood density, and graph centrality. The complete node vector will be

$$\mathbf{h}_i = \left[\underbrace{x_{i1}, \dots, x_{iM}}_{\text{protein expression}}, \underbrace{c_{i1}, \dots, c_{iT}}_{\text{cell type one-hot}}, p_i, H_i, \underbrace{C_D(i), C_B(i), C_C(i)}_{\text{derived features}} \right]$$

Edge feature captures the relationship of Euclidean distance, protein expression similarity, cell type interaction encoding, protein expression difference. Final edge vector will include

$$\mathbf{e}_{ij} = [d_{ij}, \text{sim}_{ij}, \text{interaction}_{ij}^{\text{one-hot}}, \Delta_{ij}]$$

3.3.1. Attention-Based GNN Model

Graph neural networks work on the principles of message passing, where nodes iteratively update their representation by aggregating information from neighboring nodes and updating the node's own representation.

$$\mathbf{h}_i^{(l+1)} = \text{UPDATE} \left(\mathbf{h}_i^{(l)}, \text{AGGREGATE} \left(\{\mathbf{m}_{j \rightarrow i} : j \in \mathcal{N}(i)\} \right) \right)$$

Where: $\mathbf{h}_i^{(l)}$ = node i's features at layer l $\mathcal{N}(i)$ = neighbors of node i $\mathbf{m}_{j \rightarrow i}$ = message from neighbor j to node i

Our graph learning module is based on a multi-head Graph Attention Neural Network (GAT). First, it transforms all node features through a learnable weight matrix, for each edge(i,j) compute how much the node should attend to neighbor j. For stable training and collecting a number of relationship patterns, we apply multi-head attention, in which many separate attention mechanisms function in parallel and their outputs are combined or averaged [23]. To enhance survival prediction, this enables the model to learn both local and long-range relationships in the tissue graph. Then the raw scores are normalized with softmax to convert into probabilities to get the attention weight $\alpha_{ij} = \frac{\exp(e_{ij})}{\sum_{k \in \mathcal{N}(i)} \exp(e_{ik})}$ and combine neighbor features using attention weights. Multiple attention head works in parallel and are concatenated, which results in each head learning different attention patterns.

3.4. Survival Prediction Model

The survival prediction task aims to model the relationship between the graph representation and patient survival outcome. The main model uses a regularized Cox Proportional-Hazards model to estimate patient risk score, where a higher value indicates greater risk of the event[24].

$$\text{risk}_i = \exp(f_i) \quad (1)$$

Here, the patient's features are represented by f_i , and are the learning weights. The Cox loss function is minimised in order to train the model.

$$\mathcal{L} = - \sum_{\text{observed events}} \left[\text{risk}_i - \log \sum_{\text{at-risk patients}} \exp(\text{risk}_j) \right] \quad (2)$$

We compared this to Random Survival Forest and DeepSurv (a neural network survival model). Performance is calculated by the concordance index (C-index). These States if a patient with a higher predicted risk actually experiences events sooner.

4. Conclusion

This proposed study introduces a novel exemplification for analyzing spatial proteomics data by integrating attention based graph to model to analyze better survival insights in tissue space. We aim to capture the inherent progression of cancer biology through dynamic graph models, moving from static spatial features. With future work of validation on independent cohorts and exploring the biological drivers of the identified features have the potential to provide more mechanistically informed and prognostically accurate tools for clinical prediction, ultimately guiding personalized treatment.

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